

Positive2Negative: Breaking the Information-Lossy Barrier in Self-Supervised Single Image Denoising

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Abstract

Image denoising enhances image quality, serving as a foundational technique across various computational photography applications. The obstacle to clean image acquisition in real scenarios necessitates the development of self-supervised image denoising methods only depending on noisy images, especially a single noisy image. Existing self-supervised image denoising paradigms (Noise2Noise and Noise2Void) rely heavily on information-lossy operations, such as downsampling and masking, culminating in low-quality denoising performance. In this paper, we propose a novel self-supervised single image denoising paradigm, Positive2Negative, to break the information-lossy barrier. Our paradigm involves two key steps: Renoised Data Construction (RDC) and Denoised Consistency Supervision (DCS). RDC renoises the predicted denoised image by the predicted noise to construct multiple noisy images, preserving all the information of the original image. DCS ensures consistency across the multiple denoised images, supervising the network to learn robust denoising. Our Positive2Negative paradigm achieves state-of-the-art performance in self-supervised single image denoising with significant speed improvements. The code is released to the public at <https://github.com/Li-Tong-621/P2N>.

1. Introduction

Image denoising plays a vital role in image processing to enhance visual quality [13], serving as a foundational technique across various computational photography applications [3, 39]. The widespread popularity of paired clean-noisy image datasets [1, 35] has catalyzed advancements in supervised methods, yielding remarkable performance [7, 51]. However, supervised methods fall short in

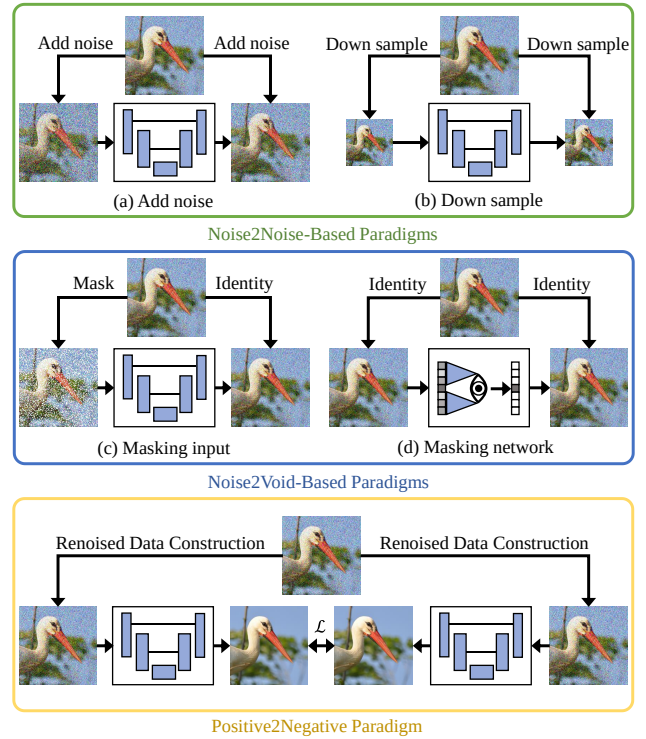


Figure 1. **Information-lossy operations in current paradigms.** The noise addition operation reduces the signal-to-noise ratio, the downsampling operation diminishes the sampling density, and the masking operation disregards crucial central pixels. Each of these operations introduces an information-lossy barrier [8, 31, 37], culminating in imprecise denoising.

dynamic scenes [18, 22], where long exposure results in misalignment and blur [22, 42], hindering the acquisition of clean images. Therefore, advancing self-supervised image denoising methods using only noisy images [25, 43], especially a single noisy image [9, 11, 24, 27], holds significant importance.

The crux of self-supervised image denoising methods

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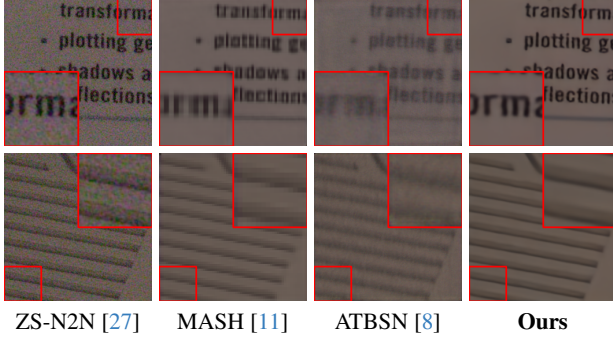


Figure 2. **The information-lossy barrier significantly compromises the image quality**, resulting in residual noise, aliasing effects or texture loss.

lies in the construction of the training data and the formulation of the supervision strategy. Current methods are divided into Noise2Noise-based paradigms and Noise2Void-based paradigms, as illustrated in Figure 1. Noise2Noise-based paradigms add additional noise to the noisy image [28, 33] or down sample the noisy image [14, 27] to approximately construct two independently noisy observations as training data. These paradigms learn denoising by supervising one observation with the denoised result of other observation. Noise2Void-based paradigms mask either the noisy image [18, 40] or the neural network [8, 21, 41] to construct masked training data. These paradigms learn denoising by supervising the denoised result of the masked central pixel with the original noisy pixel, where the prediction only relies on the surrounding pixels.

However, existing methods rely heavily on information-lossy operations, resulting in low-quality denoising performance [21, 32, 40, 41]. Specifically, the noise addition operation reduces the signal-to-noise ratio, the downsampling operation diminishes the sampling density and the masking operation disregards crucial central pixels. Each of these operations loses the information from the original noisy image [8, 31, 37], inevitably compromising the denoising ability [41, 44]. Consequently, denoised images always suffer from remaining noise, aliasing effects or texture loss [45, 46], as shown in Figure 2. As these information-lossy operations constitute the indispensable foundation of the Noise2Noise-based and Noise2Void-based paradigms, it is imperative to establish a new paradigm that transcends the Noise2Noise-based and Noise2Void-based paradigms.

In this paper, we propose the concise Positive2Negative paradigm to break the information-lossy barrier in self-supervised single image denoising. Our inspiration stems from the observation that noise distributions are approximately symmetrical and centered around zero, implying that the opposite noise also follows the distribution of the original noise. Leveraging this insight, we can construct multiple noisy images corresponding to the same clean image, which

serves as the basis of Positive2Negative. Concretely, Positive2Negative relies on two steps: Renoised Data Construction (RDC) and Denoised Consistency Supervision (DCS).

The basic idea of RDC is to construct training data. Specifically, we design a zero-mean sampling strategy to generate multi-scale Positive noise and Negative noise based on the predicted noise, which are then added back to construct Positive noisy images and Negative noisy images. The constructed data comprehensively covers the information of the given single image; thus the Positive2Negative paradigm breaks the information-lossy barrier.

The basic idea of DCS is to formulate a supervision strategy. Specifically, given the constructed multi-scale noisy images, DCS supervises the network to generate consistent denoised outputs during the training process. Furthermore, we provide a theoretical analysis showing that the Positive2Negative paradigm can learn robust denoising, through supervising the multiple denoised images.

In summary, our contributions are as follows:

- We propose a new self-supervised single image denoising paradigm, Positive2Negative, to break the information-lossy barrier.
- We propose a data construction method, which constructs multi-scale similar noisy images for training.
- We propose a denoising supervision method, which is theoretically guaranteed to learn robust denoising.
- Positive2Negative achieves SOTA performance over self-supervised single image denoising.

2. Related Work

From the training data perspective, the self-supervised image denoising methods can be classified into noisy dataset based methods and single image based methods.

2.1. Noisy Dataset based Methods

Noisy dataset based methods represent the most active research direction in image denoising. These methods learn to denoise from a given noisy dataset. Generally, a larger volume of the noisy dataset improves the denoising performance. As a result, in scenarios where only a single noisy image is available, the denoising performance of whole noisy dataset-based methods decreases significantly.

Noise2Noise [22] learns denoising by mapping one noisy observation to another independent observation of the same scene. Subsequent studies focus on generating observations of the same scene from a single observation, typically achieved through downsampling [14, 27] or by introducing additional noise [28, 33, 50, 54, 58]. After that, another elegant method Noise2Void [18] is proposed. Noise2Void [18] assumes that the noise distribution is zero-mean, the noise is spatially independent, and the signal is spatially correlated. Noise2Void [18] predicts the clean signal by leveraging surrounding noisy signals, with mask-

ing the central pixel. Various other works propose alternative masking schemes, either mask the input [11, 36, 40] or mask the network [2, 19, 20, 34, 48, 52]. Noise2Void struggles with real-world noise, which often exhibits spatial correlation. Therefore, subsequent research endeavors to break the noise spatial correlation [8, 21, 23, 41, 47, 57]. There are also some methods amalgamate frameworks established by both Noise2Noise and Noise2Void [15, 32] or get rid of frameworks established by Noise2Noise and Noise2Void [5, 16, 17, 30, 56], but these methods are usually less developed due to the complexity.

Noisy dataset based methods have high requirements for the volume of training data and fail in scenarios where only a single noisy image is available [9, 11].

2.2. Single Image based Methods

Single image based methods represent the most challenging situation for denoising and the least demanding requirements for data. These methods learn to denoise from a given single noisy image without any dependence on paired noisy-clean or noisy datasets. As a result, single image based methods are less developed due to the inherent difficulty, only gaining attention in the past two years.

DIP [38] is the earliest self-supervised image denoising method. DIP [38] learns to map random noise to a given noisy image and employs early stopping to prevent overfitting. Self2Self [36] uses dropout to create two Bernoulli-sampled observations of a noisy image. R2R [33] constructs noisy images by adding new Gaussian noise and applying reciprocal coefficients. ScoreDVI [9] performs real denoising using a pre-trained Gaussian denoiser and an estimated noise model. ZS-N2N [27] extend Noise2Noise [22] to a single image version by downsampling. MASH [11] employs masking and local pixel shuffling to break the noise spatial correlation. Recently, DMID stimulates the diffusion model for self-supervised single image denoising [24].

In this paper, we propose Positive2Negative, a new self-supervised single image denoising paradigm, to break the information-lossy barrier.

3. Positive2Negative

In this section, we first demonstrate the motivation of Positive2Negative in Section 3.1. After that, we present the two critical steps of Positive2Negative: RDC and DCS, respectively in Section 3.2 and Section 3.3. Finally, we provide an analysis and proof of Positive2Negative in Section 3.4.

3.1. Motivation

Our inspiration stems from the widely recognized assumptions that the noise distribution is zero-mean and approximately symmetrical [18, 22], which is consistent with our observation. In our observation experiment, we calculate

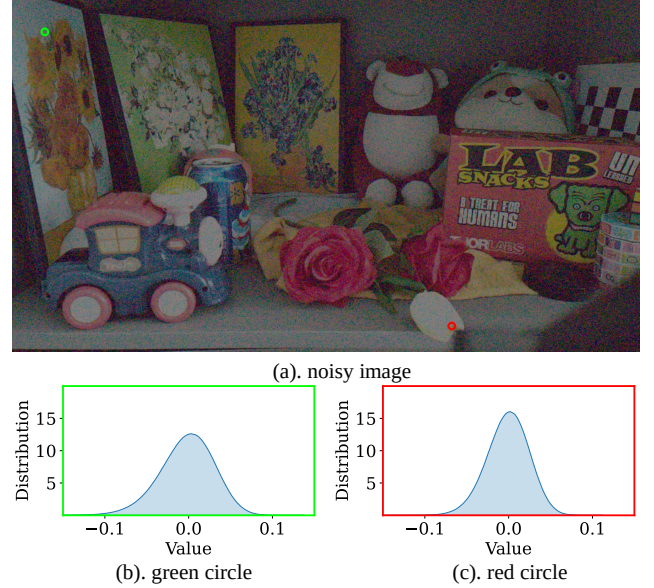


Figure 3. **Noise distribution is zero-mean and approximately symmetrical.** (a) shows a noisy image. (b) and (c) show the noise distributions, which are calculated at the center pixels marked by the green circle and the red circle in the noisy image, respectively. It is evident that the noise distribution is zero-mean and approximately symmetrical.

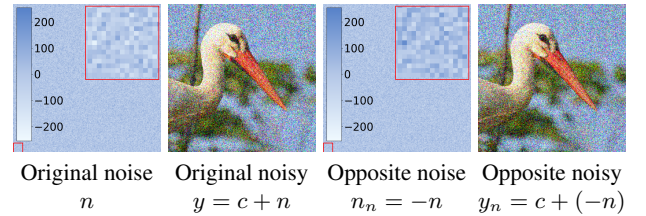


Figure 4. The opposite noise $-n$ is similar to the original noise n . The opposite noise $-n$ approximately follows the original noise’s distribution, which is zero-mean and approximately symmetrical.

the noise distribution of a fixed pixel across multiple observations of the same scene. Specifically, following SIDD [1], ELD [42] and PMN [12], we collect noisy image y and clean image c . Then we calculate the noise value n at each spatial location (i, j) . Additionally, we compute the frequency of $n_{i,j}$ cross noisy images of the same scene. The results are shown in Figure 3, which demonstrates the noise distribution is zero-mean and approximately symmetrical.

Based on this observation, an intuitive insight is that the opposite noise $-n$ also follows the same distribution as that of the original noise n . This intuitive idea inspires us to construct the opposite noise n_n and corresponding opposite noisy image y_n :

$$n_n = -n, \quad (1)$$

$$y_n = c + n_n = c - n. \quad (2)$$

The opposite noise n_n and opposite noisy image y_n are sim-

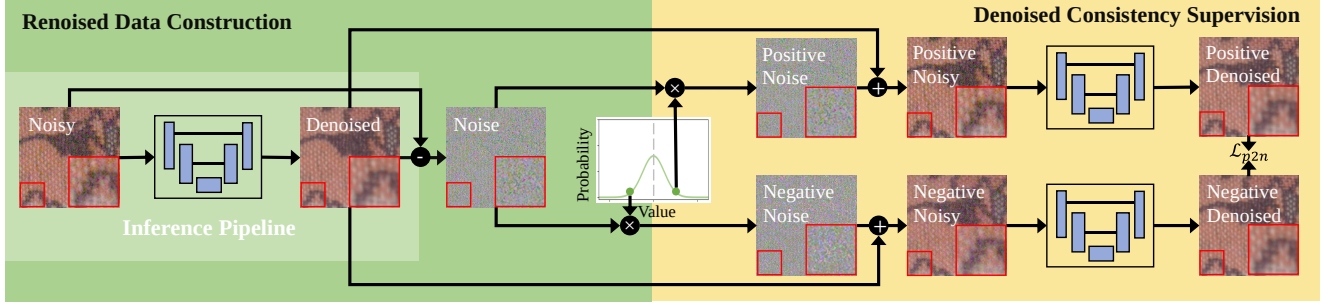


Figure 5. **Overview of the proposed self-supervised single image denoising paradigm.** For training, the Positive2Negative paradigm consists of 2 steps. The first step is Renoised Data Construction (RDC): multiply the predicted noise by positive and negative coefficients to construct multi-scale Positive noise and Negative noise, respectively. The second step is Denoised Consistency Supervision (DCS): train the network through supervising the consistency of the denoised images. For inference, denoising can be achieved in just one pass through the network. (The networks shown in the figure are the same one. In addition, for better visualization, the images within the red boxes on the bottom left have been zoomed in and are displayed in the red boxes on the bottom right.)

ilar to the original noise n and original noisy image y , respectively, as shown in Figure 4.

Based on the above observations and insights, we propose the Positive2Negative paradigm, as shown in Figure 5. Firstly, we extend our intuitive idea into a flexible and upgraded version: Renoised Data Construction (RDC), which constructs multiple positive and negative noises. Secondly, we propose the Denoised Consistency Supervision (DCS), which supervises the consistency of the predicted denoised images to learn a denoising neural network.

It is important to note that we do not require the noise distribution to be strictly zero-mean or symmetric. Although real noise generally has a zero-mean, certain cases where the noise is not strictly symmetric can be easily addressed by adjusting the norm in the loss function [22].

3.2. Renoised Data Construction (RDC)

The basic idea of RDC is to denoise the image to predict a denoised image and renoise the predicted denoised image by the predicted noise to construct multiple noisy images.

For a given noisy image $y = x + n$, where x presents the clean image and n presents the noise. RDC firstly denoises the noisy image y once through the neural network $\mathcal{F}$ parametered by θ :

$$\hat{x} = \mathcal{F}_\theta(y). \quad (3)$$

Here $\hat{x}$ is the predicted denoised image of the noisy observation y . Then we can calculate the predicted noise $\hat{n}$ within the noisy observation y :

$$\hat{n} = y - \hat{x}. \quad (4)$$

Next, RDC constructs noisy images as training data based on the predicted noise $\hat{n}$ and the predicted denoised image $\hat{x}$. As noise is zero-mean, we further design a zero-mean sampling strategy to construct new noises. Specifi-

cally, RDC multiplies the predicted noise $\hat{n}$ by both positive scale parameters σ_p and negative scale parameters $-\sigma_n$ sampled from Gaussian distribution $\mathcal{N}$, constructing multi-scale Positive noise n_p and Negative noise n_n , respectively:

$$\sigma_n, \sigma_p \sim \mathcal{N}(1, \sigma), \quad (5)$$

$$n_p = \sigma_n \hat{n}, \quad (6)$$

$$y_p = \hat{x} + n_p = \hat{x} + \sigma_n \hat{n}, \quad (7)$$

$$n_n = -\sigma_p \hat{n}, \quad (8)$$

$$y_n = \hat{x} + n_n = \hat{x} - \sigma_p \hat{n}. \quad (9)$$

3.3. Denoised Consistency Supervision (DCS)

The basic idea of DCS is that the noisy observations y_p and y_n should correspond to the same clean image. Therefore, the loss supervises the consistency of the predicted denoised images:

$$\mathcal{L}_{p2n} = \|\mathcal{F}_\theta(y_p) - \mathcal{F}_\theta(y_n)\|, \quad (10)$$

where $\|\cdot\|$ represents the norm. Considering the general nature of noise [22], we employ a gradually varying norm function $\|\cdot\|_{2 \rightarrow 1.5}$ to cover a broader range of unknown distributions:

$$\|x\|_{2 \rightarrow 1.5} = (|x| + \epsilon)^\gamma, \quad (11)$$

where $\epsilon = 10^{-8}$ and γ varies linearly from 2 to 1.5 during training. The advantage of the varying $\|\cdot\|_{2 \rightarrow 1.5}$ norm is that $\|\cdot\|_{2 \rightarrow 1.5}$ does not require fine-tuning of parameters, yet $\|\cdot\|_{2 \rightarrow 1.5}$ consistently delivers robust performance.

The noisy observations y_p and y_n contain multi-scale noise. Through DCS, the neural network $\mathcal{F}_\theta$ gradually learns to become insensitive to input noise and consistently outputs the clean signal, achieving the desired denoising performance.

Table 1. **Quantitative comparisons** on SIDD, CC, PolyU and FMDD datasets. The best and second-best results (PSNR $\uparrow$ / SSIM $\uparrow$) are marked in **red** and **blue** in self-supervised single image denoising methods. The results of noisy dataset based self-supervised methods and supervised methods are also provided, only as a reference comparison. SIDD val and SIDD ben represent the SIDD validation dataset and the SIDD online benchmark, respectively.

Category	Method	SIDD val [1]	SIDD ben [1]	CC [29]	PolyU [49]	FMDD [53]
Supervised	Unprocessing [4]	28.12 / 0.529	31.64 / 0.531	35.18 / 0.908	36.99 / 0.941	28.45 / 0.574
	MaskedDenoising [6]	28.67 / 0.604	31.99 / 0.601	33.87 / 0.930	34.56 / 0.936	29.86 / 0.643
	CLIPDenoising [10]	34.79 / 0.866	35.82 / 0.859	36.30 / 0.941	37.54 / 0.960	30.56 / 0.698
Self-Supervised (Noisy Dataset Based Methods)	Noise2Void [18]	29.35 / 0.651	27.68 / 0.668	32.27 / 0.862	33.83 / 0.873	-
	CVF-SID [30]	34.81 / 0.872	35.05 / 0.856	29.10 / 0.852	33.05 / 0.911	31.64 / 0.837
	LUD-VAE [56]	34.91 / 0.892	35.49 / 0.883	31.25 / 0.920	33.68 / 0.942	32.00 / 0.822
Self-Supervised (Single Image Based Methods)	DIP [38]	32.11 / 0.740	-	35.61 / 0.912	37.17 / 0.912	32.90 / 0.854
	PD-denoising [57]	33.97 / 0.820	-	35.85 / 0.923	37.04 / 0.940	33.01 / 0.856
	NN+denoiser [55]	-	-	36.52 / 0.943	37.66 / 0.956	32.21 / 0.831
	Self2Self [36]	29.46 / 0.595	29.51 / 0.651	37.44 / 0.948	37.52 / 0.951	30.76 / 0.695
	R2R [33]	24.59 / 0.387	-	33.43 / 0.865	36.23 / 0.931	27.17 / 0.526
	APBSN-single [21]	30.90 / 0.818	-	27.72 / 0.891	29.61 / 0.897	28.43 / 0.804
	ScoreDVI [9]	34.75 / 0.856	35.39 / 0.859	37.09 / 0.945	37.77 / 0.959	33.10 / 0.865
	ZS-N2N [27]	25.59 / 0.433	30.19 / 0.428	33.51 / 0.851	35.99 / 0.914	31.65 / 0.767
	MASH [11]	35.06 / 0.851	34.80 / 0.814	31.17 / 0.889	37.62 / 0.932	33.71 / 0.882
	ATBSN-single [8]	32.39 / 0.796	27.86 / 0.312	31.87 / 0.897	33.21 / 0.925	31.23 / 0.829
	Positive2Negative	35.24 / 0.889	35.80 / 0.875	37.92 / 0.954	38.20 / 0.961	34.00 / 0.887

3.4. Analysis

In this section, we provide an analysis and proof demonstrating how the Positive2Negative paradigm achieves denoising.

Taking the 2-norm as an example, the loss function can be expressed as:

$$\mathcal{L}_{p2n} = \mathbb{E} [\|\mathcal{F}_\theta(y_p) - \mathcal{F}_\theta(y_n)\|^2]. \quad (12)$$

Through Taylor expansion, we can approximate the predicted denoised images $\mathcal{F}_\theta(y_p)$ and $\mathcal{F}_\theta(y_n)$:

$$\mathcal{F}_\theta(y_p) = \mathcal{F}_\theta(\mathcal{F}_\theta(y)) + \sigma_p \cdot \frac{\partial \mathcal{F}_\theta(y)}{\partial y} \cdot n + \mathcal{O}(n^2), \quad (13)$$

$$\mathcal{F}_\theta(y_n) = \mathcal{F}_\theta(\mathcal{F}_\theta(y)) - \sigma_n \cdot \frac{\partial \mathcal{F}_\theta(y)}{\partial y} \cdot n + \mathcal{O}(n^2). \quad (14)$$

Substituting the approximations, we get:

$$\mathcal{F}_\theta(y_p) - \mathcal{F}_\theta(y_n) \approx (\sigma_p + \sigma_n) \cdot \frac{\partial \mathcal{F}_\theta(y)}{\partial y} \cdot n. \quad (15)$$

With further Taylor expansion, the loss function can be simplified as:

$$\mathcal{L}_{p2n} \approx (4 + 2\sigma^2) \left\| \frac{\partial \mathcal{F}_\theta(x)}{\partial x} \cdot n \right\|^2. \quad (16)$$

Through consistency loss L_{p2n} , we optimize the network $\mathcal{F}_\theta$ such that $\frac{\partial \mathcal{F}_\theta(x)}{\partial x} \cdot n$ is optimized to drive towards zero, which means that the output $\mathcal{F}_\theta(y)$ tends to:

$$\mathcal{F}_\theta(y) = \mathcal{F}_\theta(x + n), \quad (17)$$

$$= \mathcal{F}_\theta(x) + \frac{\partial \mathcal{F}_\theta(x)}{\partial x} \cdot n + \mathcal{O}(n^2), \quad (18)$$

$$\approx \mathcal{F}_\theta(x). \quad (19)$$

Ideally, the neural network $\mathcal{F}_\theta$ has learned image priors, the model output $\mathcal{F}_\theta(x)$ matches the clean signal x and we have $\mathcal{F}_\theta(x) = x$. Thus, the ultimate goal achieves:

$$\mathcal{F}_\theta(y) \approx x. \quad (20)$$

In summary, the neural network $\mathcal{F}_\theta$ gradually learns to denoise the noisy image y effectively.

4. Experiments

In this section, we first introduce the experimental settings in Section 4.1. After that, we demonstrate the performance and comparisons with other methods in Section 4.2. Finally, we conduct a detailed ablation study in Section 4.3.

4.1. Settings

In this section, we briefly introduce the experimental settings and details.

Noisy	ScoreDVI [9]	ZS-N2N [27]	MASH [11]	ATBSN-single [8]	P2N (Ours)	Reference
25.54 / 0.432	36.17 / 0.958	27.10 / 0.529	32.48 / 0.878	30.23 / 0.795	37.67 / 0.970	PSNR / SSIM
24.99 / 0.479	34.32 / 0.930	26.63 / 0.556	30.64 / 0.881	27.12 / 0.799	35.14 / 0.950	PSNR / SSIM
24.99 / 0.566	32.12 / 0.848	26.52 / 0.638	31.23 / 0.791	28.22 / 0.643	32.95 / 0.887	PSNR / SSIM

Figure 6. **Qualitative comparisons** of P2N against other self-supervised single image denoising methods in SIDD validation dataset. The denoised results of ScoreDVI are always flaky, with discontinuous lines and loss of texture. ZS-N2N does not completely denoise the images. MASH and ATBSN-single tend to either meet aliasing effect or blur the details. In summary, other methods meet aliasing effects and loss of texture, while Positive2Negative achieves both detailed texture preservation and comprehensive noise removal. In the figure, P2N represents Positive2Negative.

Datasets details. Following previous methods [9, 11], we evaluate the proposed paradigm on the widely-used datasets in self-supervised single image denoising: the SIDD validation dataset [1], the SIDD benchmark dataset [1], CC [29], PolyU [49] and FMDD [53]. The SIDD, PolyU and CC datasets consist of natural sRGB images, while the FMDD dataset consists of fluorescence microscopy images.

Implementation details. Following the default settings of self-supervised single image denoising methods [9, 11, 22, 40], we employ the same neural network architecture as Noise2Noise [22] and MASH [11] and initialize the neural network parameters with those pre-trained on DIV2K dataset with Gaussian noise [9, 55]. For training details, we employ AdamW [26] optimizer with a learning rate of 0.0001 and set scale parameter $\sigma = 0.75$, for all the experiments. No image enhancement techniques are applied.

Compared methods. Following previous methods [9, 11, 24], we mainly compare with self-supervised single image denoising methods. In addition, we also present the results of noisy dataset based self-supervised and supervised image denoising methods only as a reference comparison. All the results of the comparison methods are sourced from previous work [9, 11, 24] or evaluated by ourselves.

4.2. Comparisons

In this section, we demonstrate the performance of Positive2Negative and compare Positive2Negative with other methods both quantitatively and qualitatively.

Quantitative comparisons. The quantitative comparisons are described in Table 1. Previous self-supervised single image denoising methods have shown varying performance across different datasets, but Positive2Negative consistently achieves better results. In fact, Positive2Negative even outperforms some dataset-based and supervised methods. Notably, in the online evaluation of the SIDD benchmark dataset, we achieve an improvement of over 0.4 dB.

Qualitative comparisons. The qualitative comparisons are described in Figure 6. Clearly, the Positive2Negative paradigm achieves significant qualitative improvements by either enhancing details or effectively reducing noise. In contrast, other methods face limitations in both detail enhancement and noise removal due to the difficulty in learning denoising, resulting from the information-lossy barrier.

4.3. Ablations

In this section, we first present the training process in Section 4.3.1. Then, we conduct ablation studies on the components and hyperparameters in Section 4.3.2 and Section 4.3.3, respectively.

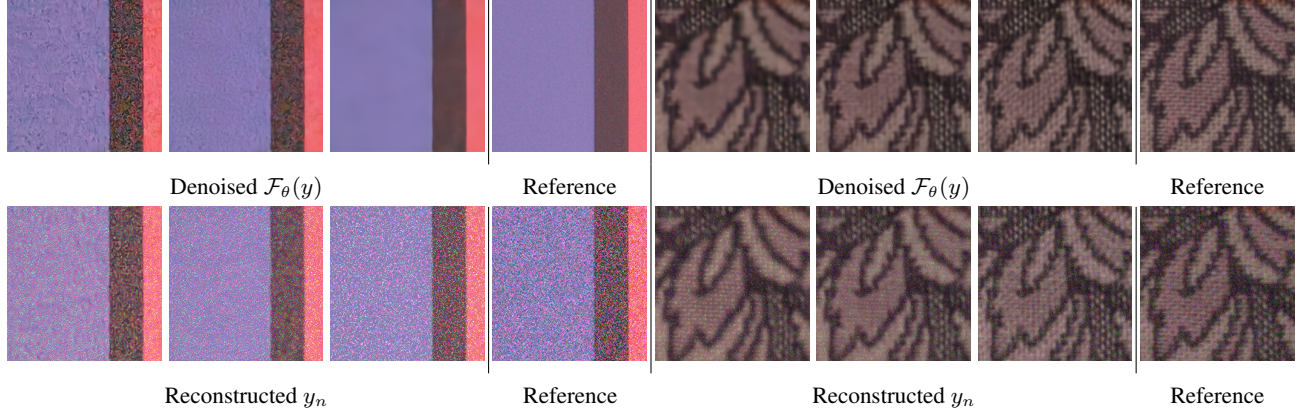


Figure 7. **Visual results** of the training process. The top row shows the denoised results, while the bottom row displays the reconstructed negative noisy images in the training process. On the far right of each set is the reference image. The denoised image and reconstructed noisy image gradually become more like the reference images.

Table 2. **Ablation of components.** Each step is crucial for Positive2Negative. N2N and N2V represent Noise2Noise and Noise2Void, respectively.

Method	CC [29]		SIDD [1]	
	PSNR	SSIM	PSNR	SSIM
Baseline	35.40	0.927	33.50	0.849
RDC+N2N	34.08	0.868	25.42	0.414
RDC+N2V	33.79	0.861	24.51	0.368
N2N+DCS	17.55	0.698	24.20	0.815
N2V+DCS	14.14	0.494	22.30	0.743
RDC+DCS	37.92	0.954	35.24	0.889

4.3.1. Ablation of training

We present visual results of the training process to illustrate the effectiveness of our Positive2Negative paradigm. As shown in Figure 7, throughout the training process, the denoised images and the reconstructed negative noisy images gradually converge towards their respective references.

4.3.2. Ablation of components

We evaluate the effectiveness of each component of the Positive2Negative paradigm on the CC and SIDD datasets. RDC constructs precise training data and DCS trains robust denoising neural networks. Both RDC and DCS are crucial for Positive2Negative. Replacing either component would result in a decline in performance, as shown in Table 2.

4.3.3. Ablation of hyperparameters

Throughout the entire training process, the only adjustable hyperparameter is the scale parameter σ , which is set to 0.75 for all experiments. As shown in Table 3, the performance remains similar for σ values ranging from 0.25 to 0.75. The scale parameter σ introduces a dynamic and wide range of noise levels, while the “Fixed” setting corresponds to $\sigma = 0$.

Table 3. **Ablation of hyperparameters.** The scale parameter is the only hyperparameter and is very robust. We fix the scale parameter $\sigma = 0.75$ for all the experiments.

Method	CC [29]		SIDD [1]	
	PSNR	SSIM	PSNR	SSIM
Fixed	37.33	0.952	35.24	0.892
$\sigma = 0.25$	37.64	0.945	35.29	0.893
$\sigma = 0.50$	37.84	0.952	35.30	0.891
$\sigma = 0.75$	37.92	0.954	35.24	0.889
$\sigma = 1.00$	37.86	0.954	35.11	0.884

5. Discussions

5.1. Distribution

Different norms are suitable for different noise distributions [22]. For simplicity, we have uniformly adopted the $\|\cdot\|_{2 \rightarrow 1.5}$ norm, as real-world noise is typically centered around a zero mean with slight fluctuations. For noise that remains nearly zero-mean, such as in PolyU [49], the $\|\cdot\|_{2 \rightarrow 1.5}$ norm and the $\|\cdot\|_2$ norm yield similar performance. For noise that exhibits slight deviations from zero mean, as in SIDD [1], the $\|\cdot\|_{2 \rightarrow 1.5}$ norm can outperform the $\|\cdot\|_2$ norm, as shown in Table 4. The slight increase in PSNR further suggests that the deviation remains minimal.

5.2. Convergence

We present the PSNR variation over iterations on SIDD dataset [1] in Figure 8. Around the 100th iteration, the Positive2Negative paradigm converges and shows minimal fluctuations thereafter. Additionally, it is worth noting that the constructed data by RDC is unsuitable for Noise2Noise and Noise2Void. This is mainly because the noise in constructed images by RDC is correlated, which would quickly collapse the training processes of the other two methods.

Table 4. **Discussion of distributions.** Different norms are suitable for different noise distributions [22]. For simplicity, we have uniformly adopted the $\|\cdot\|_{2 \rightarrow 1.5}$ norm.

Dataset	$\ \cdot\ _0$	$\ \cdot\ _1$	$\ \cdot\ _2$	$\ \cdot\ _{2 \rightarrow 1.5}$
PolyU [49]	37.88	38.02	38.20	38.20
SIDD [1]	35.12	35.26	35.20	35.24

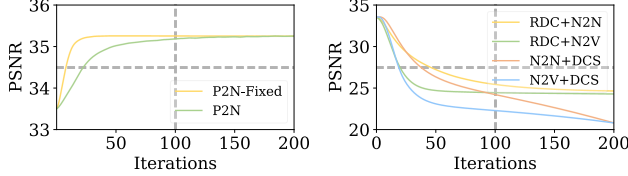


Figure 8. **Convergence comparisons.** P2N converges quickly and remains stable after convergence on SIDD. The P2N-Fixed version (where $\sigma = 0.00$), in particular, even achieves rapid convergence in approximately 10 iterations. P2N represents Positive2Negative.

5.3. Efficiency

For self-supervised single image denoising methods, computational efficiency is crucial, as each noisy image must be trained individually. Here, we demonstrate the number of training iterations, parameter, and computational cost, as detailed in Table 5. It is obvious that our paradigm achieves significant efficiency improvements.

5.4. Comparison with Noise2Noise and Noise2Void

Compared with Noise2Noise [22] and Noise2Void [18], Positive2Negative also presents an elegant paradigm. The key differences are present in the following aspects.

Firstly, for the constructed data, the synthetic data constructed by Positive2Negative breaks the information-lossy barrier of Noise2Noise and Noise2Void. Secondly, for the supervision strategy, Positive2Negative enforces consistency between denoised results, whereas Noise2Noise and Noise2Void supervise the consistency between a denoised result and a noisy image. Thirdly, from the assumption perspective, Noise2Noise, Noise2Void, and Positive2Negative all have theoretical guarantees, while Noise2Noise and Noise2Void operate under more restrictive assumptions. Noise2Noise requires two independent noisy images, while Noise2Void assumes that the noise is spatially correlated.

5.5. Differences with R2R and CVF-SID

For the constructed data, R2R employs new Gaussian noise and applies reciprocal coefficients to the noisy image, while CVF-SID employs two networks to predict three components and applies fixed manual rules. In contrast, our method employs a single network to predict noise and applies random multiscale symmetrical parameters.

Table 5. **Efficiency comparisons** under input images with dimensions of $256 \times 256 \times 3$. With similar parameters and FLOPs, P2N requires the fewest training iterations. P2N converges rapidly, and the P2N-Fixed version (where $\sigma = 0.00$) converges even just within 20 iterations. P2N represents Positive2Negative.

Method	Iterations	Params (M)	FLOPs (G)
N2N [22]	-	0.70	18.7
B2U [40]	-	1.10	18.7
Self2Self [36]	450000	1.00	9.6
R2R [33]	8000	0.56	36.7
ScoreDVI [9]	400	13.5	37.9
MASH [11]	800×3	0.99	11.4
P2N-Fixed	<20	0.99	11.4
P2N	<100	0.99	11.4

For the supervision strategy, R2R maps the denoised image to the noisy input, while CVF-SID primarily maps the first cyclic result to the noisy input and maps the second cyclic result to the first cyclic result. In contrast, our method maps the denoised image to another denoised image.

5.6. Limitation and Future Work

The primary limitation is that Positive2Negative has only been validated on the single image based self-supervised image denoising task. To prevent convergence to trivial solutions such as zero mapping or identity mapping, Positive2Negative requires training with a pre-trained Gaussian denoising model. While the single image based self-supervised denoising task allows training from a pre-trained model on Gaussian noise, the noisy dataset based self-supervised image denoising task trains from scratch. Therefore, extending the Positive2Negative paradigm to the noisy dataset based self-supervised image denoising task could be a promising direction for future work.

6. Conclusion

In this paper, we propose a self-supervised single image denoising paradigm, Positive2Negative, to break information-lossy barrier. Specifically, Positive2Negative includes two steps: RDC and DCS. Positive2Negative achieves SOTA performance compared to existing methods across various public benchmarks.

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